

SHUKRAH ADESOLA-OLABIYI

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RESEARCH

Carbon Capture Utilization and Storage (CCUS)- Fundamental Controls on Capillary Trapping of scCO₂ in Geologic Storage: Linking Pore and Reservoir Scales (DOE Sponsored)

EDUCATION

PhD in Environmental Engineering

University of Tennessee Knoxville, United States.

Jan 2025 - Dec 2030

Master of Science in Chemical and Polymer Engineering

Lagos State University, Nigeria.

Jun 2023 - Aug 2024

Bachelor of Technology in Chemical Engineering

Ladoke Akintola University of Technology, Nigeria.

Feb 2015 - Jun 2021

RESEARCH EXPERIENCE

Graduate Research Assistant

Supervisor: **Dr. Anna Herring**

Jan 2025 till date

- Designing and executing imaged and non-imaged high-pressure core-flooding experiments.
- Analyzing 3D tomography data, performing image processing, and interpreting quantitative flow and mineralization behavior.
- Participating in cross-lab data synthesis
- Engaging in pre-college outreach and mentoring, contributing to educational activities through programs like **HITES12** (UTK).

Graduate Research Assistant

Head: Engr. **Dr. A. O. Arinkoola** and **Dr. T.O Salawudeen**

Jul 2021 - Jul 2022

- Coordination and supervision of research work of final year students
- Researched on membrane separation process for wastewater treatment.
- Overseeing the laboratory research work and compilation of results
- Reading and interpretation of GC-MS and FTIR results.

WORK EXPERIENCE

Dangote Oil Refining Company

Lagos, Nigeria

Role: Process Engineer (Technical Services)

Dec 2022 – Dec 2024

- Collaborated with cross-functional teams to develop and implement safety protocols, resulting in a 30% reduction in workplace incidents.
- Supervised a team of five process engineers in the analysis of catalyst performance evaluation in hydro processing unit.
- Led process optimization initiatives, enhancing operational efficiency by implementing innovative solutions at Dangote Oil Refining Company.
- Spearheaded troubleshooting efforts, resolving complex technical issues to ensure uninterrupted refinery operations.

TEACHING EXPERIENCE

Tickle College of Engineering, UTK

United States

Graduate Teaching Assistant:

Fall semester 2025

- Assisting the course instructor in teaching computer-aided design (CAD) principles applied to civil and environmental engineering projects.
- Providing hands-on support to students during sessions on **AutoCAD 2D and 3D drafting, engineering drawing standards, and file management.**
- Supporting 40–60 undergraduate students in developing technical drawing and visualization skills.
- Grading assignments, exercises, and final design projects using standardized rubrics.

SKILLS

- **Experimental:** Core-flooding experimental design, supercritical CO₂ handling, and porous media flow characterization.
- **3D X-ray microtomography** for imaging flow and mineral reactions.
- **DRIFTS** (Diffuse Reflectance Infrared Fourier Transform Spectroscopy).
- **Modelling and Simulation:** Aspen HYSYS and Python
- **Imaging:** ImageJ/ Fiji, Webmango
- **Experimental:** Core-flooding experimental design
- **Analytical:** UV- VIS, FTIR

Others: MS Excel, MS Word, AutoCAD, Microsoft Visio

Language: English (Fluent)

Leadership, Teamwork Spirit and Good communication Skills

PUBLICATION/ CONFERENCE PRESENTATION

- **Shukrah Adesola-Olabiyi**, Christopher Allison, Daniel Enebe, Dorthe Wildenschild, Anna Herring, (September 2025). Capillary Trapping Behavior of scCO₂ During Repeated Drainage–Imbibition Cycles in Bentheimer Sandstone. (**Poster Presentation ISSE UTK Conference 2025**).
- **Shukrah Adesola-Olabiyi**, Christopher Allison, Daniel Enebe, Dorthe Wildenschild, Anna Herring, (May 2025). Capillary Trapping Behavior of scCO₂ During Repeated Drainage–Imbibition Cycles in Bentheimer Sandstone. (**Poster Presentation Interpore Conference 2025**).
- Azeez G. Akinyemi, Abass O. Alade, Akeem O. Arinkoola, **Bisola S. Olabiyi**, (May 2024). Kinetics Analysis of Crystal Violet Adsorption from Aqueous Solution into Flamboyant Pod Biochar. *Journal of the Turkish Chemical Society Section B- 1414940*. (Accepted for publication)
- **Olabiyi, B. S.**, Agbabiaka, H. O., Okweri, U. A., Oni, A. T. and Arinkoola, A.O., (November 2021). Synthesis, Evaluation and Optimization of Green Demulsifiers for Crude Oil/Water Demulsification. *2nd International Conference on Engineering and Environmental Science*. Osun State University, Nigeria. (**ICEES journal Pg 18-27**).

HONORS AND AWARDS

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| • NSF travel grant towards Interpore Conference 2025. | 2025 |
| • Best Student poster presentation award nomination, Interpore conference | 2025 |
| • Award of Appreciation, Awarded by National President , Nigeria University Engineering Students | 2021 |
| • Best student in Engineering Thermodynamics II (MEE 212) 87/100 | 2016 |
| • Best student in Transport Phenomenon IV (IV) 70/100 | 2020 |

VOLUNTEER EXPERIENCE AND LEADERSHIP

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| • Lab Manager, Herring Lab UTK | Spring Sem 2025 |
| • M.Sc Head of Class, Lagos State University | 2023-2024 |
| • Volunteer, Nigerian Universities Engineering Students Association, National Technical Conference and Exhibition (NTCE) | 2020-2021 |
| • Chairman, External Affairs at the 4th NUESA National Conference and AGM, LAUTECH | 2020-2021 |
| • Sisters' Coordinator of the Chemical Engineering Muslim Students, LAUTECH | 2019-2021 |
| • Intern lead of Warri Refining and Petrochemical Company | 2019-2020 |
| • Group Leader for Class Group Discussion | 2017-2020 |

PROFESSIONAL BODIES

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| • International Society for Porous Media (InterPore) | 2025 till date |
| • Society of Petroleum Engineers (SPE) (5124632) | 2019 till date |
| • American Institute of Chemical Engineers (AIChE) | 2019 till date |
| • Association of Professional Women Engineers (APWEN) | 2018 till date |
| • Nigeria Society of Chemical Engineer (NSChE) | 2015 till date |